

DS_Day01_26

26. MANUFACTURING — WHY IS PRODUCTION FALLING?

Industry: Automotive Manufacturing

A manufacturing plant has observed declining production output. Students receive daily machine output, downtime, shift, operator, temperature and maintenance information. They must determine when the decline started. They should identify machines responsible for the largest production losses. They should analyse downtime patterns. They should compare shifts and operators without making unsupported assumptions about individuals. They should investigate the relationship between maintenance and downtime. They should calculate Overall Equipment Effectiveness components where appropriate. The final output should identify the largest sources of productivity loss.

Data provided

Date, Machine_ID, Machine_Output, Downtime, Shift, Operator, Temperature, Maintenance_Status, Production_Volume

Student Tasks

- Clean and validate the provided dataset

Perform exploratory data analysis using suitable descriptive statistics

Create clear visualisations for the important patterns in the data

Students receive daily machine output, downtime, shift, operator, temperature and maintenance information

Determine when the decline started

Identify machines responsible for the largest production losses

Analyse downtime patterns

Compare shifts and operators without making unsupported assumptions about individuals

Investigate the relationship between maintenance and downtime

Calculate Overall Equipment Effectiveness components where appropriate

Specified Machine Learning Algorithm

Random Forest

Expected Output

5 to 7 key insights, 4 visualisations, one trained Random Forest model, model evaluation results and a practical action plan.